

Title

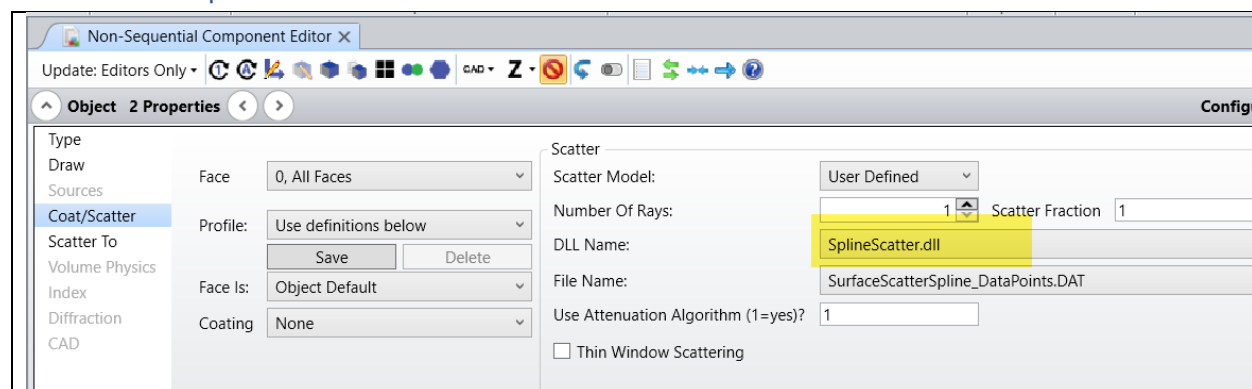
DLL_Surface_Scatter_CS_Non-sequential_Spline_scatter

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Brief Description



Overview

This Surface Scatter DLL uses an external data file with spline fit points to generate a rotationally symmetric spline-fit scatter function. Allows for quick/easy custom scatter definitions to be implemented. Spline function is defined via polar angle (with theta=0 centered on the specular ray). External data file consists only of 2 columns: polar angle, and relative radiance.

Plugin Type

Scatter DLL

Choose an item.

Language

Cpp

Updates

Date	Version	OpticStudio version	Comment
15/1/2020	1.0	19.8	Creation

Instructions

1. Installation

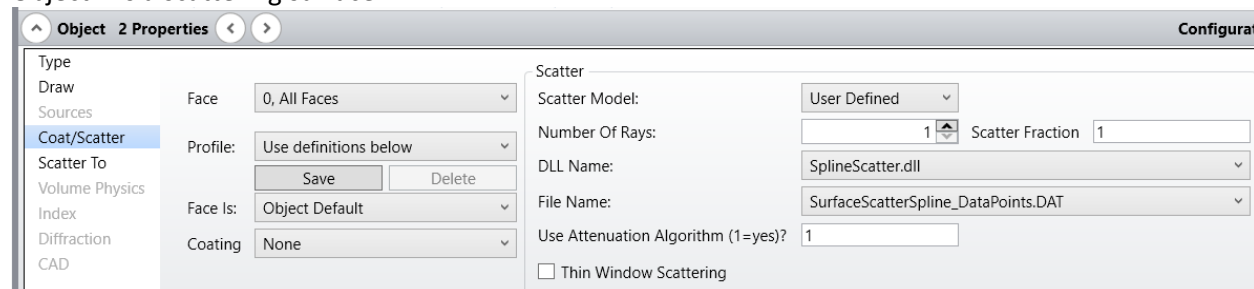
To access the DLL, copy the .dll file under {Zemax}\DLL\SurfaceScatter.

Opening the zar file will also install the dll.

2. How to use

Open the zar file.

Object 2 is a scattering surface:



To compute the spline-fit scatter function, the SplineScatter.dll attenuates the ray. By this we mean that the ray angles are chosen randomly (within allowable bounds); the output ray energies are attenuated based on spline fit.

Important points of that dll:

- The scatter function is always rotationally symmetric, and can never scatter beyond the defined data set (or to angles behind the surface itself)
- The scatter function is polarization and wavelength insensitive.
- The scatter function is sensitive to incident angle, only in that it is always centered on the specular ray.

External data file required. File formatting requirements:

- *.dat file, saved to ...{Zemax}\ScatterData\
- should only have 2 columns, space separated (no header lines, no comments, no extra blank lines anywhere including end of file)
- column 1: polar angle (measured from specular),
- column 2: relative radiance at that polar angle (expected W/sr output at that polar angle for ray tracing). The values in column 2 will not be normalized; for the attenuation algorithm to work, this means the values must be between 0 and 1
- There can be no extra lines at the start **or** end of the data file. Note that in the current implementation, there is very limited error handling for the reading of the data file; until more error handling is implemented, care should be taken in defining the data file.
- An example called SurfaceScatterSpline_DataPoints.DAT is provided.

See the source code comments for more information on how to use this DLL.

Scatter algorithm:

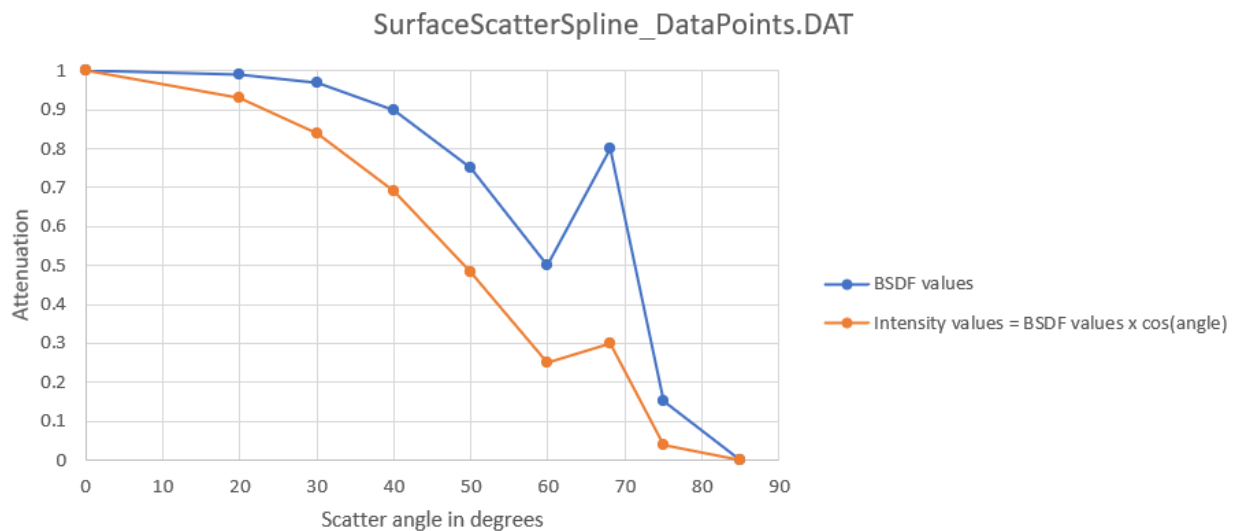
When using this DLL, the user should always run their spline data through an online natural cubic spline fitting tool, to ensure that the spline curves don't have a large section above $y=1$ (for attenuation method).

3. Validation

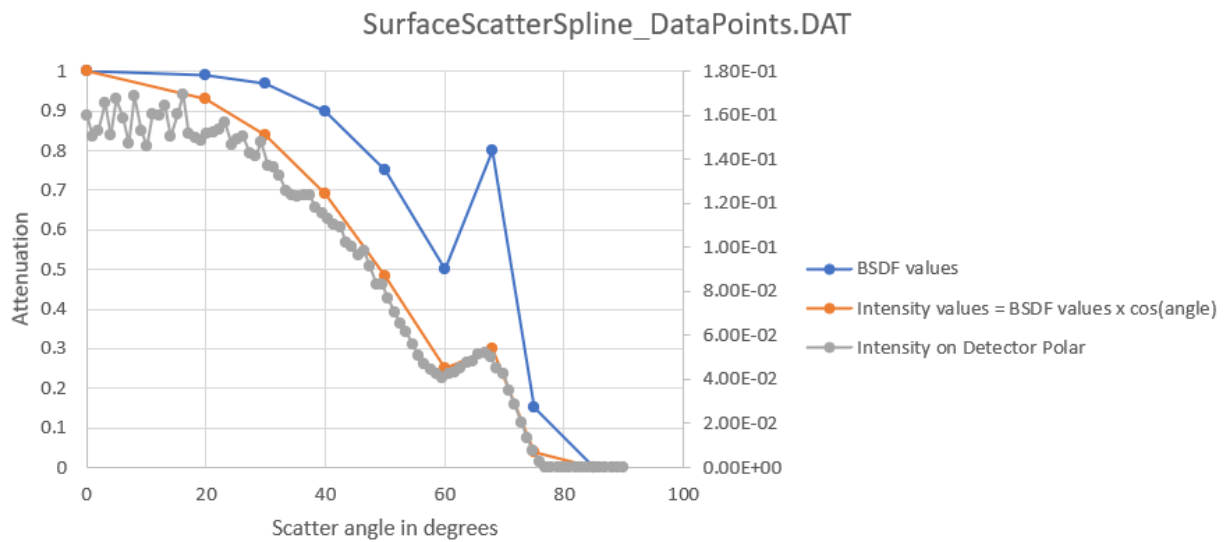
The validation is done on the attached example. The external file contains the following BSDF data:

SurfaceScatterSpline_DataPoints.DAT		
1	0	1
2	20	.99
3	30	.97
4	40	.9
5	50	.75
6	60	.5
7	68	.8
8	75	.15
9	85	0

We can plot the BSDF data (raw data from the file) as well as the intensity data ($\text{BSDF} \times \cos(\text{angle})$).



We can then compare this result with the detector polar:



The axes are not the same but the intensity recorded on the detector polar is proportional to the intensity from the file.